Project Plan

Summary

Build a sub-₹50k pipeline that scans a worn part with no drawing, recovers its original design rather than copying the wear, states how far each dimension can be trusted, and manufactures only when that bound meets tolerance. Scope it to prismatic features (bores, shafts, flats) and spur gears; that is enough to prove every claim.

Proposed poster claim: "ReNominal recovers a worn part's original design, reports how far each dimension can be trusted, and rescans or refuses when the bound misses tolerance, using a rig and pipeline under ₹50k."

The literature check below changed one thing: worn-part nominal recovery and design-intent extraction both have direct prior art, including a US patent. The defensible core is the **uncertainty gate plus uncertainty-driven targeted rescan** on a low-cost rig, backed by measured coverage of the bounds.

What the two reference projects give us

Take Orbiter's motion and pose-tracking architecture and FreeLSS's laser-stripe idea; neither does design recovery, uncertainty or manufacturing, which is where this project's contribution sits.

	Open Electronics Pi scanner (FreeLSS)	Orbiter v0.3 (Raw Mechatronics)
Method	Two 650 nm line lasers + 5 MP Pi camera, turntable, laser triangulation	2-axis rig (turntable + tilting camera arm), photogrammetry via COLMAP; v0.3 adds a stereo pair with line laser
Motion	One NEMA 17, DRV8825 at 1/32 step, up to 6,400 steps/rev	Two NEMA 17s on GT2 belts, AS5600 absolute encoders on both axes
Pose knowledge	Step count only	Records the camera's 6-DOF pose with every photo; feeds it to COLMAP as priors

	Open Electronics Pi scanner (FreeLSS)	Orbiter v0.3 (Raw Mechatronics)
Stated resolution	0.25 mm, 200 × 200 × 150 mm volume	Not stated
Known weaknesses	Calibration drift if the frame flexes; shiny/dark surfaces; misses surfaces above the camera and inside cavities	Server and firmware still model the v1 rig; v2 mechanics are print-only for now
Licence	Open source (FreeLSS on GitHub)	Code Apache-2.0, hardware CERN-OHL-S-2.0 (share-alike), docs CC-BY-4.0

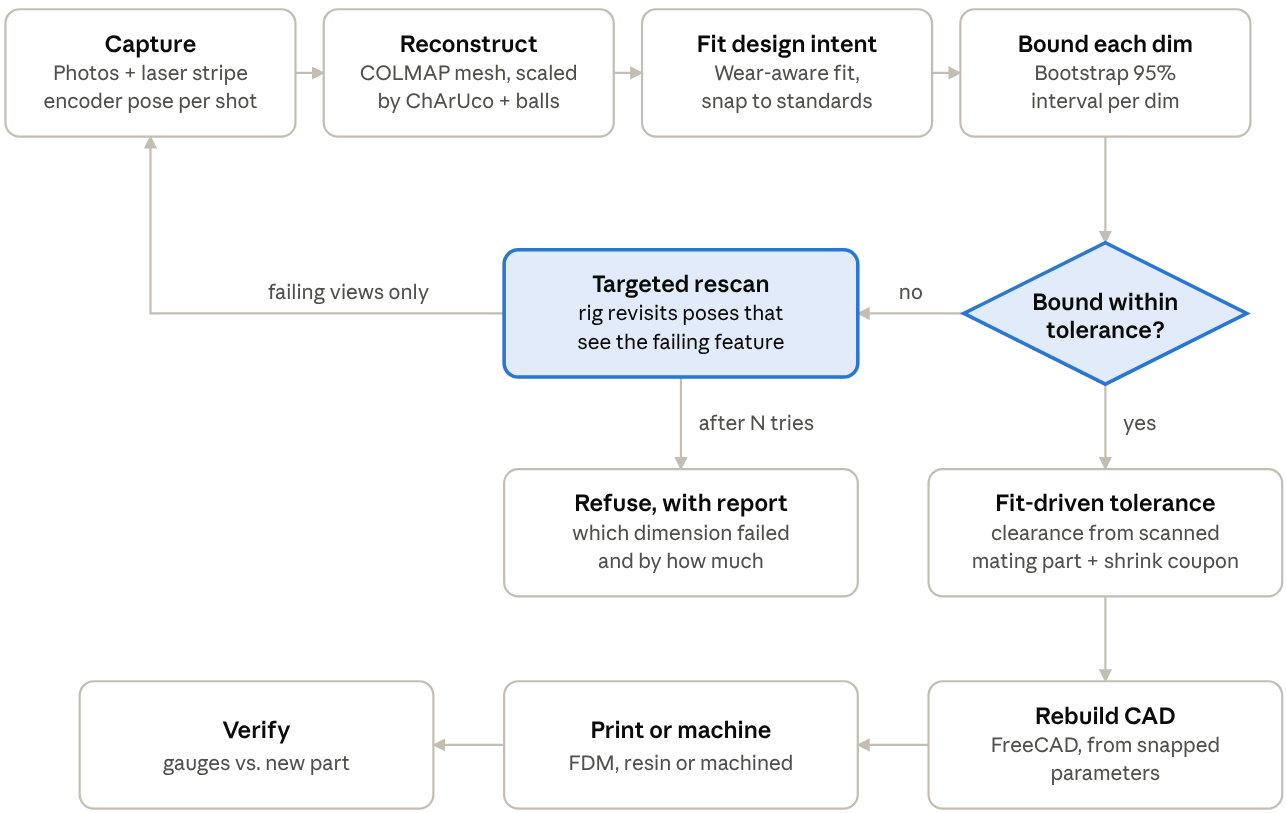
What we borrow

- From Orbiter: the turntable + tilt-arm geometry, encoder-verified poses, and pose priors into COLMAP. Known poses are what make targeted rescan (claim 1b below) possible.
- From FreeLSS: a line laser for a second, sharper measurement on functional surfaces (bores, gear flanks) where photogrammetry is weakest.
- From both: a printed calibration object. We replace it with precision steel balls and a ChArUco board, which give a metric check, not just alignment.

CERN-OHL-S means any hardware you derive from Orbiter and distribute must stay open. That is fine for an academic build, but it is one more reason the patentable part must be the method, not the rig.

System architecture

The pipeline is a closed loop: every recovered dimension carries a 95% bound, and the gate sends the rig back for more data on exactly the feature that fails, or refuses the part.



Highlighted: the novel loop, the gate (claim 1a) and targeted rescan (claim 1b)

ReNominal pipeline · capture to verified part, with the gate and rescan loop

The top row is standard reverse engineering; the highlighted gate and rescan are what make it a measurement system rather than a scan-to-CAD tool.

Novelty and patentability

Every single technique here has prior art; what is plausibly new is closing the loop — the per-dimension uncertainty bound decides whether the rig rescans, refuses or manufactures. Lead with that, and treat design-intent recovery as supporting, not headline.

Ranked claims against what the search found

#	Claim	Closest prior art found	What is still yours	Proof on the poster	Patent angle
1a	Manufacture-or-don't gate with per-dimension uncertainty bounds	Sobh et al. built geometric entities with uncertainty models from noisy sensor data for re-manufacturing (Utah, 1993); Konecki et al. estimate nominal tolerance fields for worn spur gears, but with micrometers, not scans (Applied Sciences 2024)	Bounds computed per fitted parameter from the scan itself, checked against tolerance automatically, on a low-cost rig	Coverage: share of 95% bounds that contain the measured truth, across all parts	Weak alone (concept is old); strong as part of 1b
1b	**Uncertainty-driven targeted rescan** (new suggestion)	Not found in this search; generic "next-best-view" planning exists in robotics, so search it before claiming	When a feature's bound misses tolerance, the rig moves to the encoder poses that view that feature and rescans until the bound passes or a limit is hit	Bound width before vs. after targeted rescan; time saved vs. full rescan	**Best candidate.** Ties software to a physical rig and a measurable technical effect
2	Design-intent recovery instead of scan-copy	Li & Li fit a nominal model to worn parts after excluding damaged regions (RCIM 2017); US7814441B2 claims computing design intent from scan mesh regions; Shi & Bai matched a worn involute gear to a standard CAD model (MEMS 2012)	A low-cost, open workflow; the scan-copy vs. recovered comparison on real parts	Side-by-side error maps: scan-copy vs. recovered, against the new reference part	Low — directly anticipated

#	Claim	Closest prior art found	What is still yours	Proof on the poster	Patent angle
2b	Wear-aware fitting: on a one-direction gear, fit the involute to the **unloaded (coast) flank** and snap module/pressure angle to standard series	Konecki et al. treat the nominal module as fixed, not random; bevel-gear work recovers macro-parameters from scans (Gear Technology)	Using the loaded/unloaded flank asymmetry to weight points, with a likelihood over standard values	Recovered module and pressure angle vs. the new gear	Possible dependent claim under 1b
3	Fit-driven tolerancing from the scanned mating part	Tolerance approximation for RE exists (Design Society)	Clearance set from the mating part's scanned bound plus a measured printer-shrink coupon	First-print fit rate vs. nominal-size baseline	Dependent claim at most
4	Measured accuracy-vs-scan-quality curve	Many single-scanner accuracy studies; the 2025 worn-part study compared a handheld and a stationary scanner but copied the worn shape rather than recovering it (MSPE 2025)	Recovery error (not scan error) vs. scan noise across DIY, entry-level and bureau scans	One curve, with bound coverage shown at each level	None — this is a research contribution, publish it
5	Open validation set of worn/reference pairs	None found in this search	The dataset itself	Published with code and ground truth	None — publishing it defeats patenting the same content

Not novel; present as enabling tools: photogrammetry, the rig, laser triangulation, primitive fitting, ICP alignment, 3D printing.

Patent notes for an Indian student team

These are general points, not legal advice; take them to your institute's IPR cell early.

- **Disclosure kills novelty.** A poster, demo, report upload or GitHub push is public. File a provisional application before any of these, or keep claim 1b out of public material until it is filed.

- **Software alone is hard to patent in India.** Section 3(k) of the Patents Act excludes computer programs per se. Claim 1b is framed as a measurement apparatus that changes its own acquisition, which is a stronger position than a pure algorithm.
- **Open hardware and open data are prior art too.** If you publish the dataset (claim 5) and the rig design, they cannot be claimed later. Decide what to file and what to publish in the same meeting.
- **Run the 30-minute search before filing.** Add "next best view uncertainty reverse engineering" and "adaptive rescan tolerance" on Google Patents and Scholar to the searches already suggested.

Bill of materials

The recommended build comes to about ₹41,000 including three bureau scans, leaving roughly ₹9,000 of margin under ₹50k. Prices are approximate Indian retail estimates (Robu, Amazon.in, local markets), not quotes; check before ordering. Borrow anything marked "lab".

Item	Qty	Approx. ₹	Notes
Rig — motion			
NEMA 17 stepper motor	2	1,400	Azimuth turntable + camera tilt arm
TMC2209 stepper driver	2	700	Quieter and smoother than DRV8825
ESP32 dev board	1	400	Orbiter firmware runs on ESP32
AS5600 magnetic encoder + magnet	2	400	Absolute angle on both axes; the basis for targeted rescan
GT2 belts + 20T/large pulleys	1 set	700	Large pulley can be printed
Turntable bearing (6008 pair or lazy-susan)	1 set	500	
2020 aluminium extrusion + brackets	~2 m	1,500	Stiff frame; flex was FreeLSS's main error source
12 V 5 A supply + buck converter	1	700	

Item	Qty	Approx. ₹	Notes
PLA/PETG for printed parts	1 kg	1,000	
Fasteners, wiring, limit switches	—	600	
Imaging			
Raspberry Pi 5 (4 GB) + SD + PSU	1	7,500	Camera control; or use a laptop + USB camera
Raspberry Pi HQ Camera	1	5,000	Fixed focus, no phone-style processing
16 mm C-mount lens	1	2,500	Fills frame on 20–80 mm parts
650 nm line laser, 5 mW	2	600	Second measurement on bores and flanks
LED panels + diffuser / light tent	1 set	2,000	Even, shadow-free light for photogrammetry
Calibration and ground truth			
ChArUco board (printed, glued to glass or ACP)	1	500	Camera intrinsics + metric scale
Precision steel balls, 3 sizes (grade 10–25)	~10	500	Known-diameter artefacts for accuracy checks
Developer spray (dye-penetrant type)	2 cans	1,000	Matte coat for shiny/black parts
Digital vernier, 0–25 mm micrometer, pin gauges, gear tooth vernier	—	lab	Ground-truth measurement
New reference parts (one per worn part)	~10	3,000	Ground truth for recovery error
Manufacturing and comparison			
PETG/nylon filament for test prints	1 kg	1,500	FDM on lab printer
Resin print service for tight-fit parts	~5 parts	2,500	Better than FDM for gear teeth

Item	Qty	Approx. ₹	Notes
Service-bureau / entry-scanner scans	3 parts	6,000	Upper end of the accuracy curve (claim 4)
Recommended total		**≈ 41,200**	

Cheaper fallback (≈ ₹10,000): a smartphone on a 3D-printed arm, one stepper turntable, no Pi or HQ camera, no bureau scans. It still demonstrates claims 1a and 2, but with wider bounds and no targeted rescan.

Do not buy an entry-level 3D scanner (₹35k+). Borrow one from a makerspace or college lab for the comparison scans, or use the bureau budget.

Software stack

All free and open source; the only code you write from scratch is the fitting, uncertainty, gate and rescan planner (roughly 1,500–2,500 lines of Python).

Stage	Tool	Your work
Rig control	Orbiter firmware (ESP32) + FastAPI server	Adapt to your motors; add a "go to poses" endpoint for targeted rescan
Camera calibration and scale	OpenCV ChArUco	Intrinsics once; scale check on steel balls every session
Photogrammetry	COLMAP (with Orbiter's pose priors) or Meshroom	Settings only
Laser stripe	OpenCV stripe extraction, FreeLSS as reference	Triangulate stripes on selected features
Cleaning, alignment, comparison	Open3D, CloudCompare	ICP to reference; cloud-to-mesh error maps
Feature fitting	Open3D RANSAC + SciPy least squares	Cylinders, planes; involute profile fit for spur gears
Uncertainty	NumPy bootstrap over scan points (and over repeat scans)	95% interval per fitted parameter

Stage	Tool	Your work
Gate and rescan planner	Python	Compare interval to tolerance; choose encoder poses that see the failing feature
CAD rebuild	FreeCAD Python API (+ FreeCAD Gears workbench)	Parametric model from fitted, snapped values
Manufacture	OrcaSlicer / PrusaSlicer	Printer-shrink coupon to set hole and shaft compensation
Report	Jupyter or a small web page	One-page verdict per part: each dimension, its bound, pass/fail

Validation plan

The single number that proves claim 1 is **coverage**: across all parts and features, about 95% of the stated 95% bounds should contain the measured true value. Everything else supports it.

Test parts (8–10 worn/new pairs)

Pick parts that wear out often and whose new replacement is cheap — the new part is your ground truth.

Part	Where to get it	Features to recover
Mixer-grinder coupler (jar or motor side)	Any appliance repair shop; new ones ₹50–150	Splines/lugs, bore, outer diameter
Nylon spur gears (pair)	Appliance, printer or toy gearboxes	Module, tooth count, pressure angle, bore
GT2 or similar timing pulley	Old 3D printer or CNC	Tooth profile, bore, flange
Plain bushing / sleeve	Fan, motor or fixture	ID, OD, length
Knob or handle with D-shaft bore	Appliance	D-flat bore, outer form

Part	Where to get it	Features to recover
Bracket with hole pattern	Fixture or machine	Hole positions, hole diameters, flatness

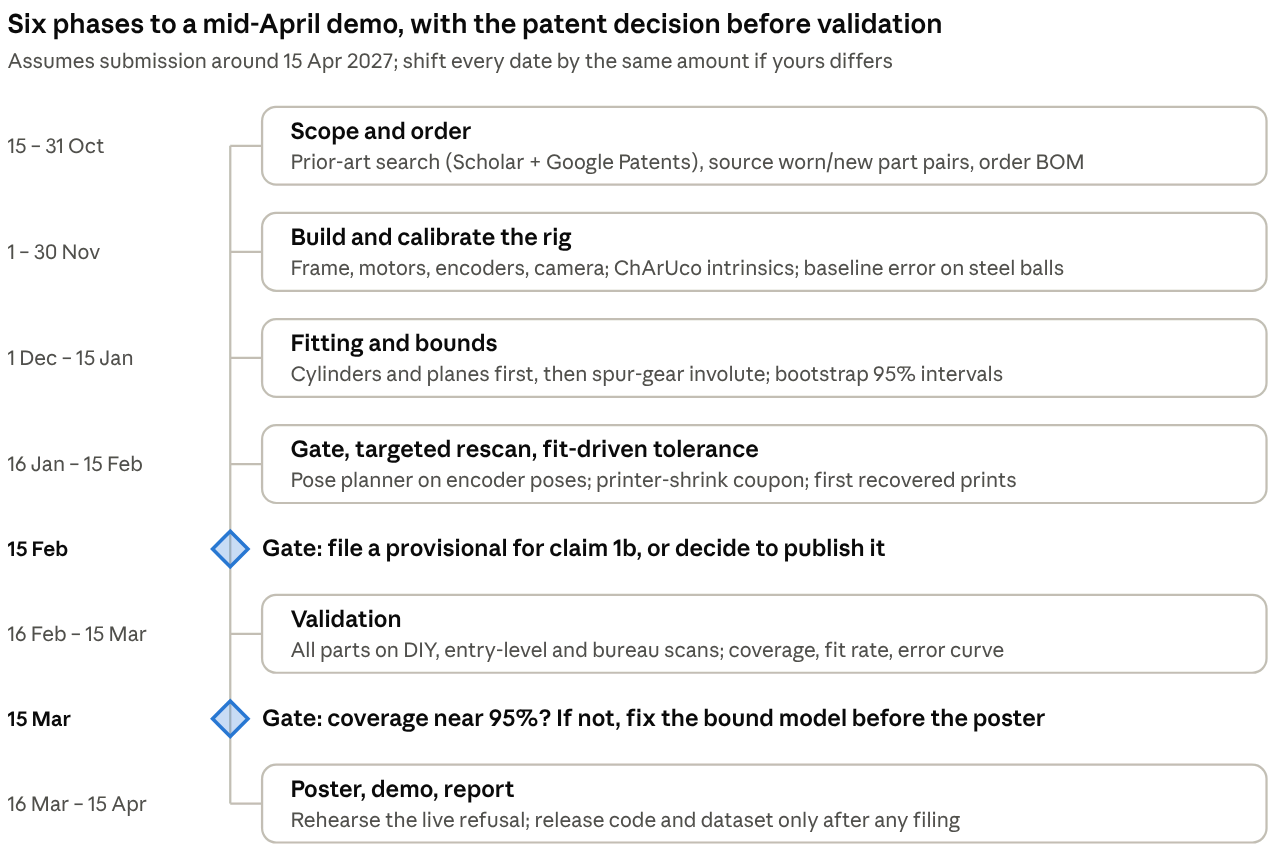
Metrics

Metric	Claim it proves	How
Per-feature recovery error	2	Recovered value minus value measured on the new part (micrometer, pin gauge, gear tooth vernier; CMM if the college has one)
Scan-copy error	2 (baseline)	Same features measured on the raw scan mesh
Bound coverage	1a	Share of 95% intervals containing the true value
Gate confusion matrix	1a	Accepted-and-fits, accepted-but-fails, refused-would-fit, refused-correctly
Bound width before / after targeted rescan	1b	Same part, same tolerance; also count extra photos and minutes
First-print fit rate	3	Fit-driven vs. nominal-size clearance, same printer, same mating part
Recovery error vs. scan noise	4	DIY rig, entry-level scanner and bureau scan of the same parts

Repeat each scan three times per setup so the bootstrap reflects repeatability, not just point noise. Fix the gate tolerance per process: about ±0.2 mm for FDM, ±0.05 mm for resin, stated on the poster.

Timeline

Six months, with the patent decision on 15 Feb so nothing about targeted rescan goes public before it.



Project roadmap · 6 phases, 2 gates, Oct 2026 to Apr 2027

Open question: what is your actual submission date and review schedule? The plan assumes mid-April 2027.

Risks and mitigations

The biggest risk is that DIY scans are too noisy for gear teeth; the gate turns that into a result ("refused, here is why") rather than a failure.

Risk	Likelihood	Mitigation
Photogrammetry too coarse for small gear teeth	High	Laser stripe on flanks; snap module to the standard series, which is discrete, so moderate noise still picks the right value
Shiny, black or translucent parts fail to scan	High	Developer spray; choose test parts accordingly

Risk	Likelihood	Mitigation
FDM cannot hold the recovered tolerance	Medium	Gate tolerance set per process; resin service for gears; recovered CAD is still valid for machining
No reliable ground truth	Medium	Buy a new part for every worn part; ask the metrology lab for CMM time
Orbiter v2 hardware not yet supported by its software	Medium	Build the simpler v1-style rig or write minimal stepper firmware; only pose accuracy matters
Scope creep into freeform parts	Medium	Prismatic features and spur gears only; everything else is "future work"
Public disclosure before patent filing	Medium	Decide filing by the gate date in the timeline; keep claim 1b out of public repos until then
Close prior art surfaces for claim 1b	Medium	Narrow to the low-cost validated system and the coverage result; publish instead

Demo and poster plan

Evaluators judge the demo, so build it around one live moment: the rig refuses a part, rescans only the weak feature, then passes it.

1. **Table layout:** worn part, scan-copy print, recovered print and new reference part in a row, each with its gauge reading on a card.
2. **Live gate:** scan a gear with a deliberately short capture. The report shows the bore passing and the module bound failing tolerance. The rig turns to the flank poses, rescans, and the bound narrows to a pass.
3. **Fit check:** the recovered part slides onto its mating shaft; the nominal-size print does not.
4. **Poster exhibits:** the architecture diagram, the coverage result (one number, large), the recovery-error vs. scan-quality curve, and the first-print fit rate.
5. **Tools & Techniques panel:** photogrammetry, laser stripe, rig, printing — labelled as enabling tools, not innovation.

The poster sentence in the Summary only states what the demo shows. A live demo of targeted rescan discloses it, so file any provisional application before the first public showing.

Sources

- An open source 3D Scanner made with Raspberry Pi — Open Electronics
- Orbiter — Lab-Kit Pan-Tilt Camera Rig — Raw Mechatronics
- Konecki et al., Evaluating the Accuracy of the RE Process of Worn, Non-Standard Spur Gears, Applied Sciences 2024
- Li & Li, An integrated approach of RE aided remanufacturing for worn components, RCIM 2017
- US7814441B2 — Identifying original design intents using 3D scan data
- Shi & Bai, RE technology of laser cladding repairing gears, MEMS 2012
- Sobh et al., Analysis of Tolerance for Manufacturing Geometric Objects from Sense Data, Univ. of Utah 1993
- Dimensional Tolerance Approximation for RE Applications — Design Society
- The Use of RE for the Development of a Model Using CAD, MSPE 2025
- 3D Scan-Based RE of Differential Bevel Gears — Gear Technology

Prices in the bill of materials and the Indian patent notes are from general knowledge, not from these sources.